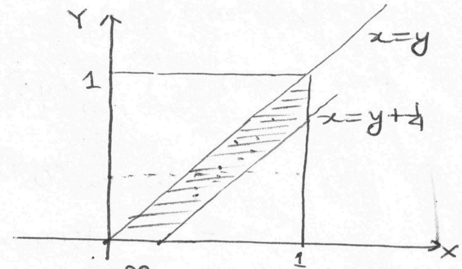
Example: An electronic device is designed to switch house lights on and off at random times after it has been activated. Assume that it is designed in such a way that it will be switched on and off exactly once in a one hour period. Let Y denote the time at which the lights are turned on and X the time at which they are turned off. Assume the joint density fun for (X,Y) is

$$f_{X,Y}(x,y) = \begin{cases} 8xy, & 0<y<x<1 \\ 0, & \text{e.w.} \end{cases}$$

(i) Find the probability that the lights are switched on within $\frac{1}{2}$ hour after being activated and then switched off again within 15 minutes.

$$= P\left(Y<\tfrac{1}{2},\ X<Y+\tfrac{1}{4}\right)$$

$$= \int_0^{\frac{1}{2}} \int_y^{y+\frac{1}{4}} 8xy\,dx\,dy = \frac{11}{96}$$



(ii) Find the probability that the lights will be switched off within 45 minutes of the system being activated given that they were switched on 10 minutes after the system was activated.

$$P\left(X\leq\tfrac{3}{4}\,\middle|\,Y=\tfrac{1}{6}\right) = \int_{\frac{1}{6}}^{\frac{3}{4}} \frac{72}{35}x\,dx = \frac{77}{140} = \frac{11}{20}.$$

$$f_Y(y) = \int_y^1 8xy\,dx = 4y(1-y^2), \quad 0<y<1$$

$$f_{X|Y=y}(x|y) = \frac{8xy}{4y(1-y^2)} = \frac{2x}{1-y^2}, \quad y<x<1,\ 0<y<1$$

So $f_{X|Y=\frac{1}{6}}(x) = \frac{72}{35}x, \quad \frac{1}{6}<x<1.$

(iii) Find the expected time that the lights will be turned off again given that they were turned 10 minutes after the system was activated.

$$E\left(X|Y=\tfrac{1}{6}\right) = \int_{\frac{1}{6}}^{1} x f_{X|Y=\frac{1}{6}}(x)\,dx = \int_{\frac{1}{6}}^{1} \frac{72}{35}x^2\,dx = \frac{43}{63}.$$

(iv) Find $\rho_{X,Y}$

$$E(XY) = \int_0^1 \int_0^x 8x^2y^2\,dy\,dx = \frac{4}{9} \qquad Cov(X,Y) = \frac{4}{9} - \frac{4}{5}\cdot\frac{8}{15}$$

$f_X(x) = 4x^3,\ 0<x<1, \quad E(X) = \frac{4}{5},\ E(X^2) = \frac{2}{3},\ V(X) = \frac{2}{75}$

$f_Y(y) = 4y(1-y^2),\ 0<y<1, \quad E(Y) = \frac{8}{15},\ E(Y^2) = \frac{1}{3},\ V(Y) = \frac{11}{225}$

$\rho \cong 0.4924$